

Retail Banking Financial Performance & Benchmarking Dashboard

Power BI | DAX | Power Query | Excel | Financial Analytics

1. Project Overview

The **Retail Banking Financial Performance & Benchmarking Dashboard** is an interactive Power BI analytical solution developed to analyse and compare the financial performance of selected Indian banks across multiple financial years and key banking metrics.

The project brings together financial information obtained from publicly available bank financial reports and disclosures. The information was organized and standardized into a structured analytical dataset before being transformed and incorporated into a Power BI data model.

The completed solution provides an interactive environment for examining financial performance from multiple perspectives, including historical trends, year-over-year changes, peer comparisons, bank rankings, benchmark analysis and performance gaps.

The dashboard is organized into three analytical pages:

- **Executive Overview** — provides a high-level summary of selected financial KPIs and their performance.
- **Financial Deep Dive** — provides detailed analysis of individual financial metrics, including historical trends and peer comparisons.
- **Bank Benchmarking & Performance Insights** — focuses on bank ranking, benchmark comparison, performance gaps and dynamically generated recommendations.

The dashboard supports the following analytical capabilities:

- Historical KPI analysis
- Year-over-year performance analysis
- Peer bank comparison
- Bank ranking
- Project-level benchmark analysis
- Performance gap analysis
- Category-level analysis
- Dynamic KPI selection
- Interactive filtering
- Dynamic analytical insights
- Dynamic recommendations

The solution is designed as an **analytical reporting and decision-support project**. It demonstrates how financial information can be transformed into a structured analytical model and presented through an interactive business intelligence solution.

Project Coverage

Area	Coverage
Industry	Retail Banking / Banking
Geography	Selected Indian Banks
Analysis Type	Financial Performance & Benchmarking
Reporting Tool	Microsoft Power BI
Data Transformation	Power Query
Analytical Language	DAX
Data Organization	Microsoft Excel
Time Dimension	Multiple Financial Years
Comparison	Selected Peer Banks
Benchmark	Calculated from Banks Included in Dataset
Dashboard Pages	3

Banks Included

The project dataset includes the following selected banks:

1. Axis Bank
2. HDFC Bank
3. ICICI Bank
4. Kotak Mahindra Bank
5. State Bank of India

The exact financial KPIs available for analysis depend on the information incorporated into the project dataset.

Analytical Flow

The overall analytical flow of the project can be represented as:

Bank Financial Reports

↓

Data Organization & Standardization

↓

Data Consolidation

↓

Power Query Transformation

↓

Power BI Data Model

↓

DAX Analytical Measures

↓

Interactive Dashboard

↓

Performance Insights & Benchmarking

2. Business Problem

Financial information published by banks is generally available through annual reports, financial statements and other public disclosures. However, the information is often presented in different formats, terminology and reporting structures.

When financial performance needs to be compared across several banks and multiple financial years, manually reviewing individual reports can make the process time-consuming and difficult to analyse consistently.

The project addresses this problem by organizing selected financial information into a structured analytical dataset and developing an interactive Power BI dashboard for comparative analysis.

The analytical solution is designed to help answer questions such as:

- How has a selected financial KPI changed over time?
- What is the year-over-year movement of a selected KPI?
- How does one bank compare with its peer banks?
- Which bank performs strongest under a selected analytical context?
- What is the relative rank of a selected bank?
- How does a selected bank compare with the calculated benchmark?
- Is the selected bank performing above or below the benchmark?

- What is the performance gap?
- Which categories or KPIs may require further analysis?

The project therefore focuses on transforming dispersed financial information into a structured, interactive analytical environment.

3. Project Objectives

The primary objectives of the project are:

1. **Consolidate financial information** for selected Indian banks across multiple financial years.
 2. **Standardize financial metrics** into a consistent structure suitable for comparative analysis.
 3. **Organize and prepare the data** so that bank-level financial information can be analysed using common dimensions.
 4. **Build a structured Power BI data model** using fact and dimension tables.
 5. **Develop DAX measures** for KPI analysis, year-over-year calculations, rankings and benchmarking.
 6. **Enable dynamic KPI analysis** through user-selected metrics.
 7. **Enable interactive filtering** by bank, financial year, category and KPI.
 8. **Compare bank performance** across the selected peer group.
 9. **Calculate relative benchmarks** using the banks included in the project dataset.
 10. **Identify performance gaps** between selected bank performance and the calculated benchmark.
 11. **Provide dynamic analytical recommendations** based on the selected analytical context.
 12. **Present the analysis through an interactive three-page Power BI dashboard.**
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4. Scope of Analysis

The project focuses on comparative financial performance analysis of selected Indian banks.

The analysis covers:

- Multiple financial years
- Multiple banking categories

- Multiple financial KPIs
- Selected peer banks
- Historical KPI trends
- Year-over-year changes
- Peer comparison
- Bank ranking
- Benchmark comparison
- Performance gap analysis
- Category-level analysis
- Dynamic recommendations

Banks Included

The project dataset includes:

Bank

Axis Bank

HDFC Bank

ICICI Bank

Kotak Mahindra Bank

State Bank of India

The exact KPIs available for analysis depend on the financial information incorporated into the project dataset.

Analytical Scope

The dashboard is designed primarily for **historical and comparative analysis**.

It does not attempt to:

- Forecast future financial performance
- Predict stock prices
- Provide investment advice
- Replace official financial statements
- Function as a live banking reporting system

The calculated benchmark is also limited to the banks included in the project dataset.

5. Data Sources

The financial information used in the project was collected from publicly available bank financial reports and disclosures.

Relevant financial metrics were identified from these reports and organized into a structured dataset for comparative analysis.

The source information was not incorporated directly in its original report format. Instead, the relevant information was organized and standardized before being incorporated into the Power BI analytical model.

The overall data flow was:

Bank Financial Reports

↓

Data Organization & Standardization

↓

Data Consolidation

↓

Power Query Transformation

↓

Power BI Data Model

↓

DAX Analysis

↓

Interactive Dashboard

The use of publicly available financial information allows the dashboard to focus on analytical modelling and comparative financial performance rather than confidential banking data.

6. Data Preparation

Data preparation was an important stage of the project because financial information from different banks may vary in terminology, presentation and structure.

The objective of preparation was to create a consistent structure across banks, financial years, categories and KPIs.

Key Preparation Activities

The preparation process included:

- Organizing bank-level financial information
- Standardizing bank names

- Standardizing financial year information
- Standardizing KPI names
- Structuring metric values
- Consolidating bank-level datasets
- Preparing data for Power BI
- Maintaining consistent units for comparable metrics
- Validating the organized information

Power Query

Power Query was used for data transformation and consolidation before analysis in Power BI.

The transformation process helped prepare the data in a consistent format suitable for loading into the analytical model.

Excel

Excel was used during the data organization, validation and preparation process.

The prepared information was structured so that the resulting data could be consolidated and incorporated into the Power BI model.

7. Data Model

The Power BI solution uses a structured analytical data model consisting of fact tables, dimension tables and supporting tables.

The central analytical table is:

Fact_Bank_Metrics

This table contains the consolidated financial metrics used throughout the dashboard.

The model also contains dimension and supporting tables including:

Table	Purpose
Fact_Bank_Metrics	Consolidated financial metrics
Dim_Bank	Bank information and bank selection
Dim_Date	Financial-year and time analysis
Dim_Metrics	KPI and metric information
KPI_Details	KPI-level analytical presentation

Table	Purpose
Executive_Insights	Dynamic insight generation
Bank Selector	Selected-bank analysis and comparison
Measures 2	Supporting analytical measures
Variance_Steps	Variance-related analytical calculations
Waterfall Steps	Variance and presentation logic

The model also contains bank-specific source tables that were used during data preparation and modelling.

7.1 Fact Table

The central analytical table is Fact_Bank_Metrics.

It stores the consolidated financial metric values used by the dashboard.

The fact structure allows financial values to be analysed across dimensions such as:

- Bank
- Financial Year
- Category
- KPI

This structure provides a common analytical base for calculations and visualizations across the report.

Using a consolidated fact structure also allows measures to operate consistently across different banks and KPIs.

7.2 Dimension Tables

Dim_Bank

The bank dimension provides bank-level information used for:

- Bank filtering
- Bank selection
- Peer comparison
- Ranking analysis

Dim_Date

The date dimension supports financial-year analysis.

It also contains an ordering mechanism that allows financial years to be treated chronologically for calculations such as previous-year analysis.

Dim_Metrics

The metrics dimension contains KPI-related information used to organize and filter financial metrics.

It supports dynamic KPI selection and allows the dashboard to work with multiple metrics within the same analytical framework.

8. DAX & Analytical Logic

DAX was used to develop the analytical calculations required for the dashboard.

The measures support:

- KPI calculations
- Historical analysis
- Year-over-year analysis
- Bank ranking
- Benchmarking
- Performance gap analysis
- Dynamic insights
- Dynamic recommendations

The calculations respond to the active filter context selected by the user.

This means that changing the bank, financial year, category or KPI can change the relevant analytical outputs.

8.1 Total Value

The foundational measure is:

Total Value =

`SUM(Fact_Bank_Metrics[Metric_Value])`

This measure calculates the total metric value within the current filter context.

It serves as a foundation for several other calculations used throughout the report.

8.2 Previous Year Value

The previous-year calculation uses the financial-year ordering available in the date dimension.

The logic identifies the currently selected financial year and retrieves the corresponding value from the preceding financial year.

This provides the basis for year-over-year comparison.

The approach allows previous-year analysis to respond dynamically to the selected analytical context rather than relying on a fixed year.

8.3 Year-over-Year Change

The absolute year-over-year change is calculated as:

YoY Change =

[Total Value] - [Previous Year Value]

This measure represents the difference between the selected financial year's value and the corresponding previous financial year's value.

A positive result indicates an increase relative to the previous year, while a negative result indicates a decrease.

8.4 YoY Percentage

In addition to the absolute change, the dashboard calculates the percentage change between the current and previous financial year.

The percentage measure allows users to understand the relative scale of movement rather than only the absolute difference.

For example, two KPIs may show the same absolute increase while representing very different percentage movements because their underlying values are different.

8.5 Selected KPI Value

The dashboard uses dynamic KPI selection so that users can select a financial metric and view its corresponding value within the selected analytical context.

The selected KPI calculation retrieves the value associated with the chosen KPI.

This allows the same dashboard structure to support multiple financial metrics without requiring a separate dashboard page for each KPI.

8.6 Bank Ranking

The model calculates the relative position of a selected bank within the peer group under the selected analytical context.

The resulting ranking supports the **Selected Bank Rank** card on the benchmarking page.

The ranking can change when the analytical context changes, such as when users select a different:

- Financial year
- Category
- KPI
- Bank comparison context

This allows ranking to remain relevant to the current analysis.

8.7 Benchmark Calculation

The dashboard calculates a benchmark using the banks included in the analytical dataset.

The benchmark acts as a reference point for evaluating selected bank performance.

The benchmark is calculated within the relevant analytical context, allowing it to change according to the selected metric, category and financial year.

Important Interpretation

The benchmark should be interpreted as a **project-level peer benchmark**.

It represents the comparison base created from the banks included in the project dataset and **does not represent the entire Indian banking industry**.

8.8 Performance Gap

The dashboard calculates the difference between selected bank performance and the relevant benchmark.

The result is presented through the:

Performance Gap vs Industry Average (%)

visual.

The performance gap helps identify whether the selected bank is relatively above or below the calculated benchmark.

This provides an additional layer of analysis beyond simply displaying the bank's absolute financial value.

8.9 Dynamic Recommendations

The dashboard contains recommendation logic that interprets selected bank performance relative to the benchmark.

The recommendation changes according to the selected analytical context rather than displaying a fixed statement.

The purpose of this logic is to translate numerical comparison into a concise analytical interpretation.

The recommendations should therefore be viewed as **dashboard-generated analytical guidance for further investigation**, rather than financial advice or an independent assessment of a bank.

9. Dashboard Design

The Power BI report is organized into three analytical pages.

Each page addresses a different level of financial analysis.

9.1 Executive Overview

The **Executive Overview** provides a high-level summary of financial performance.

It is designed as the starting point for users who want to quickly understand the selected KPI and its movement across banks and financial years.

Filters

The page contains slicers for:

- Category
- Bank
- Financial Year
- KPI

Key Visuals

Peer Bank Comparison

Provides a comparative view of the selected KPI across the peer banks included in the dataset.

KPI Trend

Shows the historical movement of the selected KPI across financial years.

Executive Insight

Displays a dynamically generated textual interpretation based on the selected analytical context.

KPI Card

Displays the selected KPI.

Selected KPI Value

Displays the value associated with the selected KPI and filter context.

YOY %

Shows the year-over-year percentage movement.

YOY Change

Shows the absolute year-over-year change.

Purpose

The Executive Overview provides a concise starting point for identifying major financial movements before moving into more detailed analysis.

9.2 Financial Deep Dive

The **Financial Deep Dive** page provides a more detailed examination of individual financial KPIs.

Filters

The page includes:

- Bank
- Financial Year
- Category
- KPI

Key Visuals

KPI Cards

The page displays:

- Selected KPI
- Selected KPI Value
- YOY %

Trend Line Chart

Shows the historical movement of the selected financial metric.

Clustered Bar Chart

Provides a comparative visual representation of financial performance.

Peer Bank KPI Summary

Provides KPI performance information across the selected peer banks.

KPI Details

Provides a more detailed KPI-level view of the selected analytical context.

YoY Change

Displays the absolute change between the selected financial year and the previous year.

KPI Insight

Provides a dynamically generated interpretation of the selected KPI.

Purpose

The Financial Deep Dive page allows users to move beyond high-level reporting and examine individual metrics in greater detail.

9.3 Bank Benchmarking & Performance Insights

The **Bank Benchmarking & Performance Insights** page focuses on relative bank performance.

It combines bank comparison, ranking, benchmarking and recommendation logic.

Filters

The page includes:

- Financial Year
- Category
- Compare Bank

KPI Cards

Best Performing Bank

Identifies the strongest-performing bank under the selected analytical context.

Selected Bank Rank

Shows the relative position of the selected bank within the peer group.

Average Benchmark Score

Provides a benchmark-oriented performance measure.

Industry Benchmark

Displays the calculated benchmark used for comparison.

Category-Level Recommendation

Provides a recommendation based on the selected category and analytical context.

Recommendation for Selected Bank

Provides a dynamic recommendation for the selected bank based on its relative performance.

Comparative Visuals

Peer Bank KPI Comparison

Provides a detailed comparison of banks within the selected context.

Performance Gap vs Industry Average (%)

Shows the relative difference between bank performance and the calculated benchmark.

Units

The dashboard explicitly identifies the principal units used:

Units: Balance Sheet values are in ₹ Crores. Shareholder metrics are in ₹ per share.

This clarification helps prevent incorrect interpretation when switching between different financial metrics.

10. Analytical Framework

The dashboard follows a layered analytical framework.

This structure allows the analysis to move from historical performance to peer comparison, benchmarking and interpretation.

Layer 1 — Historical Analysis

The first layer examines financial metrics across financial years.

Key questions include:

- Is the metric increasing or decreasing?
- What is the year-over-year movement?
- What is the historical trend?
- Is the direction of performance consistent over time?

The KPI trend visuals provide the primary interface for this analysis.

Layer 2 — Peer Comparison

The second layer compares the selected bank with other banks included in the dataset.

Key questions include:

- Which bank has the highest value?
- Which bank has the lowest value?
- How does the selected bank compare with peers?
- How does the relative position change across different KPIs or years?

The peer comparison visuals provide this comparative perspective.

Layer 3 — Benchmark Analysis

The third layer evaluates selected bank performance against the calculated benchmark.

Key questions include:

- Is the selected bank above or below the benchmark?
- What is the performance gap?
- How does the selected bank compare with the peer-group reference point?

This layer provides context to the selected bank's absolute financial value.

Layer 4 — Ranking

The ranking layer determines the relative position of banks within the selected analytical context.

Ranking provides another way of interpreting peer performance because a bank's position can be evaluated without relying only on absolute values.

Layer 5 — Insight & Recommendation

The final layer converts numerical comparisons into concise analytical interpretations.

Dynamic insights and recommendations are generated according to the selected analytical context.

This helps users move from:

Data → Comparison → Interpretation

while keeping the underlying numerical analysis visible.

11. Interactive Analysis

A major feature of the dashboard is its filter-driven design.

Users can modify the analytical context using slicers such as:

- Bank
- Financial Year
- Category
- KPI

- Compare Bank

The calculations and visuals respond to the selected filter context.

For example, changing the selected bank or KPI can update relevant outputs such as:

- Selected KPI Value
- YOY %
- YOY Change
- Bank Rank
- Benchmark comparison
- Performance gap
- Dynamic insight
- Recommendation

This design allows users to investigate multiple combinations of banks, KPIs, categories and financial years within the same Power BI report.

Instead of creating separate reports for every combination, the dashboard uses dynamic calculations and filter context to provide multiple analytical views through the same interface.

12. Technical Considerations

12.1 Metric Standardization

Bank reports may use different terminology, layouts and presentation formats for similar or related financial information.

Standardization was therefore an important part of preparing the dataset for comparative analysis.

The organized dataset uses consistent bank names, financial-year information, KPI names and metric structures.

12.2 Financial Year Handling

Financial-year analysis requires consistent chronological ordering.

The date dimension contains an ordering mechanism that supports chronological analysis and previous-year calculations.

This enables the dashboard to distinguish the current financial year from its preceding year for YoY analysis.

12.3 Dynamic KPI Analysis

The dashboard uses dynamic KPI selection instead of requiring a separate dashboard structure for each financial metric.

This allows the same analytical framework to display different KPIs based on user selection.

Dynamic KPI analysis reduces visual duplication while maintaining a consistent user experience.

12.4 Benchmark Comparability

Benchmarking requires the selected banks and metrics to be interpreted consistently.

The dashboard applies benchmark calculations within the selected analytical context and uses the standardized project dataset as the comparison base.

Because the benchmark is derived from the project dataset, it should be interpreted as a peer-group reference rather than an external industry-wide benchmark.

12.5 Units of Measurement

Financial metrics may use different units of measurement.

The dashboard explicitly identifies the primary units used in the analysis:

- **Balance Sheet values — ₹ Crores**
- **Shareholder metrics — ₹ per share**

This distinction is important when users switch between different KPIs.

12.6 Filter Context

The analytical measures are designed to operate within the active Power BI filter context.

This allows calculations to respond to user selections rather than relying entirely on static values.

The use of filter-driven measures is particularly important for:

- KPI selection
- Financial-year analysis
- Bank comparison
- Ranking
- Benchmarking
- Performance gap analysis
- Dynamic recommendations

13. Key Analytical Outputs

The completed dashboard provides the following analytical outputs:

KPI Performance

Displays financial KPI values according to the selected analytical context.

Historical Trends

Shows how selected financial metrics have changed across financial years.

Year-over-Year Analysis

Provides both absolute and percentage-based comparisons with the previous financial year.

Peer Comparison

Allows the performance of selected banks to be compared with other banks in the dataset.

Bank Ranking

Shows the relative position of a selected bank within the peer group.

Benchmark Analysis

Provides a calculated reference point based on the banks included in the project dataset.

Performance Gap Analysis

Shows the relative difference between selected bank performance and the calculated benchmark.

Category-Level Analysis

Allows users to examine financial performance according to different analytical categories.

Dynamic Insights

Provides textual interpretations based on the selected analytical context.

Dynamic Recommendations

Provides dashboard-generated recommendations based on relative performance and benchmark comparison.

14. Project Workflow & Maintainability

The project follows a structured workflow from financial data organization through transformation, modelling and dashboard development.

Workflow

Step 1 — Financial Reports

Relevant financial information was collected from publicly available bank reports and disclosures.

Step 2 — Data Organization

Required financial metrics were organized into a structured format.

Step 3 — Data Cleaning & Standardization

Bank names, financial years, KPI names and metric structures were standardized.

Step 4 — Data Consolidation

Bank-level information was consolidated into a common analytical structure.

Step 5 — Power Query Transformation

Power Query was used to transform and prepare the consolidated data.

Step 6 — Power BI Data Model

The prepared data was incorporated into the Power BI model using fact and dimension structures.

Step 7 — DAX Measures

DAX measures were developed for KPI values, YoY analysis, ranking, benchmarking, performance gaps and recommendations.

Step 8 — Dashboard Development

The analytical outputs were presented through three interactive Power BI pages.

Maintainability

The structured model and reusable DAX measures provide a framework that can be updated when compatible additional data becomes available.

New data would need to follow the established structure and standardization approach to maintain consistency with the existing analytical model.

15. Tools & Technologies

Tool / Technology	Usage
Power BI	Data modelling, visualization, DAX calculations and dashboard development
DAX	KPI calculations, YoY analysis, ranking, benchmarking and recommendations

Tool / Technology	Usage
Power Query	Data transformation, consolidation and preparation
Excel	Data organization, validation and preparation

Power BI

Power BI serves as the primary analytical and visualization environment.

It was used for:

- Data modelling
- Measure development
- Interactive visualizations
- Slicers and filtering
- Dashboard layout
- Analytical reporting

DAX

DAX was used to create dynamic analytical measures.

The calculations support:

- Current KPI values
- Previous-year values
- YoY change
- YoY percentage
- Ranking
- Benchmarking
- Performance gaps
- Insights
- Recommendations

Power Query

Power Query was used to transform and consolidate the prepared financial information before it was loaded into the Power BI analytical model.

Excel

Excel was used during data organization, validation and preparation.

16. Project Structure

The recommended GitHub repository structure is:

Retail-Banking-Financial-Performance-Analysis/

```
|
|
|— README.md
|
|— Dashboard/
|   |— Retail_Banking_Dashboard.pbix
|   |
|   |
|   |— Dashboard_Screenshots/
|       |— executive-overview.png
|       |— financial-deep-dive.png
|       |— bank-benchmarking.png
|
|— Data/
|   |— README.md
|
|— Documentation/
    |— Retail_Banking_Project_Documentation.pdf
```

Repository Components

Root README.md

Provides a concise introduction to the project, including:

- Project overview
- Business problem
- Objectives
- Tools
- Dashboard preview
- Key analytical outputs
- Repository structure

Dashboard Folder

Contains the Power BI .pbix file and dashboard screenshots.

Data Folder

Contains information about the project dataset and its preparation process.

The Data/README.md explains the source, preparation and limitations of the data.

Documentation Folder

Contains the detailed project documentation PDF.

The PDF provides the complete technical and analytical explanation of the project.

17. Limitations

17.1 Source Data Dependency

The analysis depends on the accuracy, completeness and consistency of information available in publicly published financial reports.

Differences in reporting formats and disclosures can affect the availability and comparability of individual metrics.

17.2 Metric Availability

Not every financial metric is necessarily available in an identical form across all banks and financial years.

Therefore, comparisons should be interpreted within the specific KPI and analytical context selected in the dashboard.

17.3 Benchmark Scope

The calculated benchmark represents the banks included in this project dataset.

It should not be interpreted as a benchmark for the entire Indian banking industry.

The term "Industry Benchmark" displayed within the dashboard refers to the calculated benchmark used by the project and should therefore be understood in the context of the available peer dataset.

17.4 Historical Analysis

The dashboard focuses on historical and comparative financial analysis.

It does not include:

- Financial forecasting
- Predictive modelling

- Future performance projections
 - Stock-price prediction
-

17.5 Production Use

The project is an analytical portfolio solution and is not intended to function as a production banking reporting system.

It does not replace official financial statements, regulatory reporting systems or audited financial disclosures.

18. Conclusion

The **Retail Banking Financial Performance & Benchmarking Dashboard** combines financial data organization, data transformation, structured data modelling, DAX-based analytical calculations and interactive Power BI visualization into a single analytical solution.

The project provides a structured framework for examining:

- Financial KPI performance
- Historical trends
- Year-over-year changes
- Peer bank comparisons
- Bank rankings
- Benchmark comparisons
- Performance gaps
- Category-level performance
- Dynamic insights
- Analytical recommendations

The three-page Power BI report allows users to move progressively from a high-level **Executive Overview**, to detailed **Financial Deep Dive** analysis, and finally to **Bank Benchmarking & Performance Insights**.

The underlying model uses a consolidated fact structure together with bank, date and metric dimensions, enabling calculations to respond to the selected analytical context.

DAX measures provide the logic required for KPI analysis, YoY calculations, ranking, benchmarking and performance interpretation, while Power Query supports the transformation and consolidation of the prepared data.

Overall, the project demonstrates an end-to-end analytical workflow:

Financial Information → Data Preparation → Data Transformation → Data Modelling → DAX Analysis → Interactive Visualization → Comparative Insights

The resulting dashboard provides an interactive framework for exploring the financial performance of selected banks across financial years, KPIs and peer groups.